

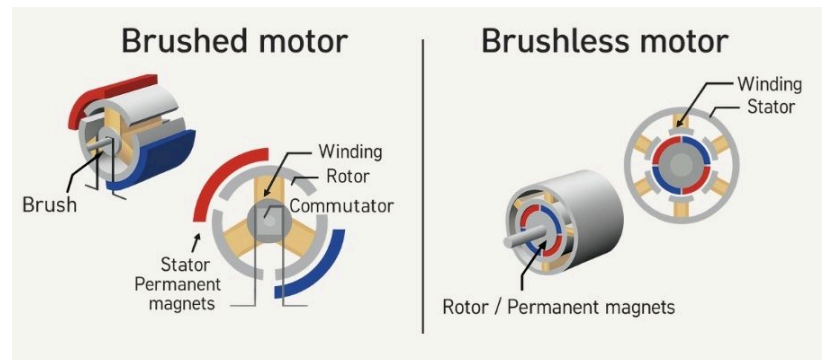
# 1. Background

## 1.1 Brushed vs. Brushless DC Motors

DC motors are typically composed of two concentric magnetic-field generating components: a circular array of permanent magnets for a fixed magnetic field, and a circular series of wound copper coils, which can act as variable magnets depending on the electrical current passed through them.

Brushless motors are distinct from brushed DC motors in their commutation method, or how current is switched in the copper motor windings to generate a moving magnetic field against the permanent magnets.

In brushed motors, the wound coils are typically found in the rotor (the rotating component), where a direct-contact brush armature supplies a switching current to induce rotation against a permanent magnet stator, which remains stationary. In brushless motors, the wound coils are the unmoving stator, which requires an external circuit board to induce a spinning magnetic field, which generates resultant movement in the permanent-magnet rotor.



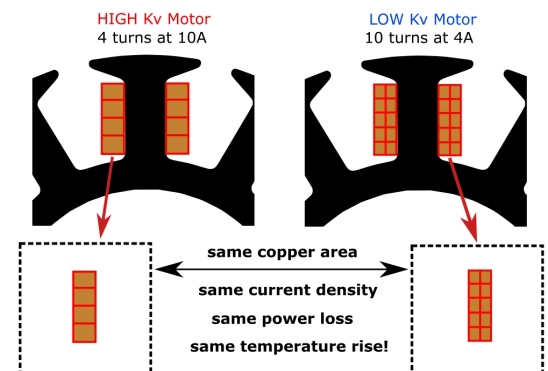
Brushed motors tend to experience higher noise and wear due to the direct contact of the brush armature against the spinning rotor, though it remains relatively low cost and uncomplex in its hardware requirements. However, brushless motors thus experience longer lifespans and efficiencies, and lower noise from lack of direct-contact parts, but requires a more complex control scheme for the wound-coil stator from an electronic speed controller (ESC).

## 1.2 BLDC Motor Configurations

The characteristics of a BLDC motor can be adjusted according to its construction to support different applications of the motor.

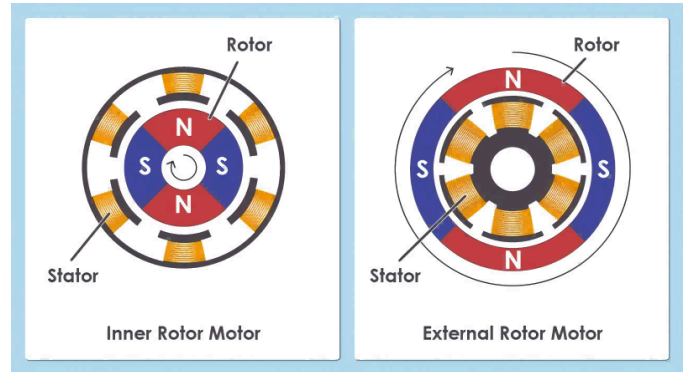
### Winding Count and KV

The KV value of a motor refers to its constant velocity rating, or RPM per volt. Generally, fewer windings of copper on the stator core's teeth will result in a higher KV, and thus higher motor speed with less torque, particularly used in high speed drone motor applications. More windings of the same copper area will result in a lower KV rating, generally used to support lower speed, higher-torque applications such as in robotics or heavy-lift drones.



### Inrunner vs Outrunner Motors

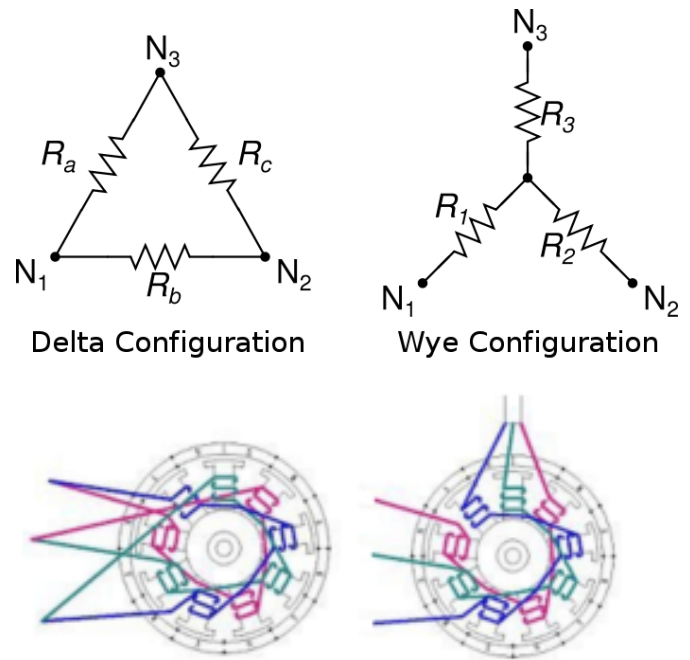
The orientation of the stator and rotor can impact the performance of a BLDC motor. As the permanent magnets always act as the spinning rotor in a BLDC, it can exhibit higher torques when set as the outer shell of the motor, and higher speed in inrunner motors.



### Delta vs. Star/Wye Configurations

Unlike a single-phase brushed motor, most BLDCs involve three distinct motor phases that are wired in a 120 degree orientation.

The pattern in which the three-phase coil windings are wired on a motor can adjust its performance. Star configurations are generally suitable for low-speed applications, as its efficiency is higher at a low RPM. Delta configurations provide better torque efficiency at a high speed and high power. Star configurations involve a neutral point, and less phase current, while delta configurations involve higher phase current and thus higher heat concentration.



### 1.3 Motor Control Theory

The switching of currents across the three motor phases in brushless DC motors can be controlled by a variety of methods, such as trapezoidal (six step), sinusoidal, and field-oriented control (FOC). This is performed by an electronic speed controller with MOSFET gates performing high-frequency PWM.

Compared to six-step and sinusoidal commutation, FOC provides high motor efficiency, low torque ripple, and low noise, compared to trapezoidal and sinusoidal commutation.

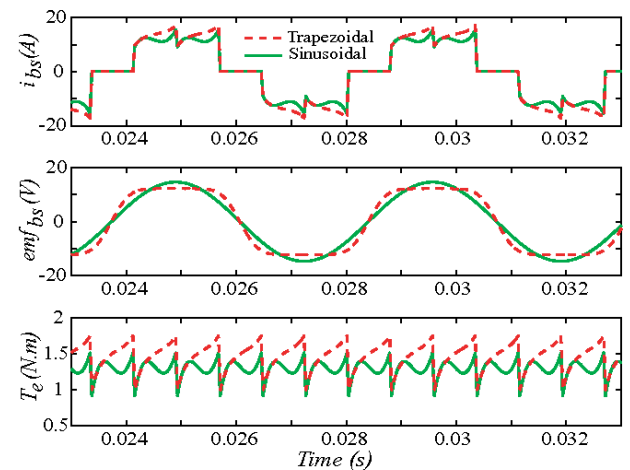
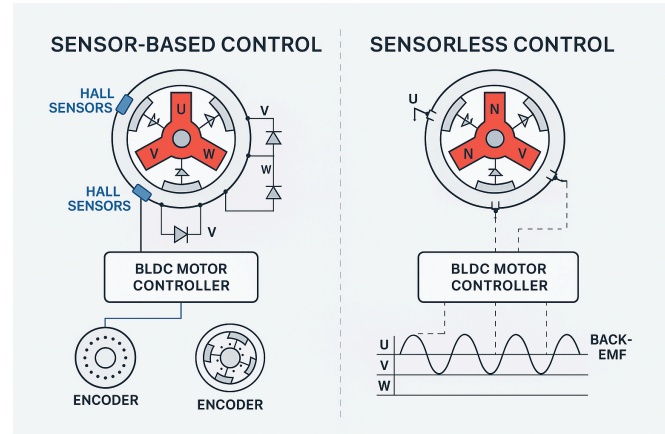


Figure 3. Steady-state waveforms of phase current, back-EMF, and developed electromagnetic torque as predicted by various models.

## Position Feedback

As with all control schemes, some form of a feedback loop must be placed to compare control inputs with a measured outcome, and subsequent corrections enacted by the control loop can be performed.

Whenever motor movement is generated, a force opposing the direction of motion is generated, known as back-emf. Since only two of the three motor phases are powered at one time, the back-emf in the floating phase can be read to estimate position without an encoder. This is the basis of voltage-based sensing, in less complicated boards or sensorless commutation.



In more complicated schemes, sensor feedback to detect the motor's actual position will ensure a more accurate control scheme. In BLDC motors, this is performed using either Hall-effect sensors or an absolute magnetic encoder.

## 1.4 Field Oriented Control

FOC is an evolution of sinusoidal commutation, performing vector control of magnetic flux, current, and voltage in a rotating d-q reference frame. For the most relevant force production, the goal is to maximize the torque vector  $q$  (quadrature), while minimizing the outward-facing flux vector  $d$  using PI controllers.

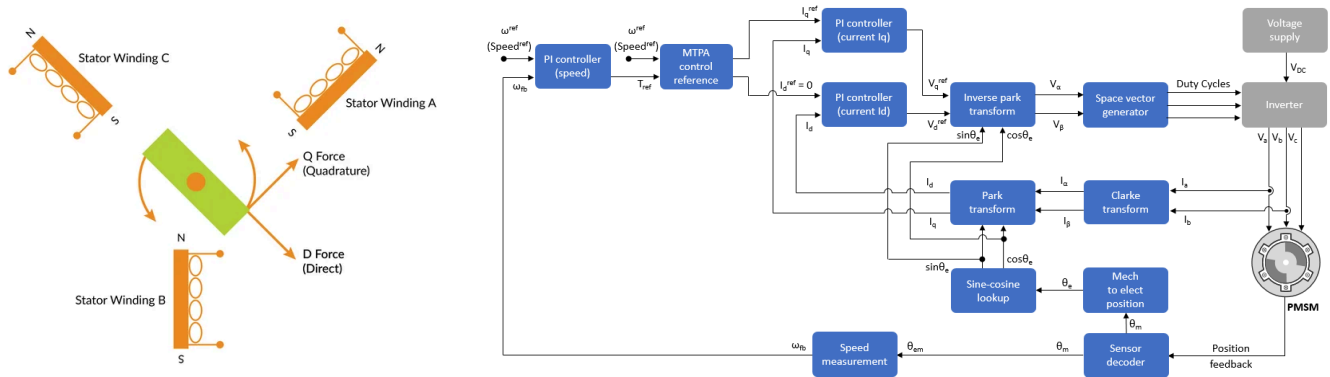
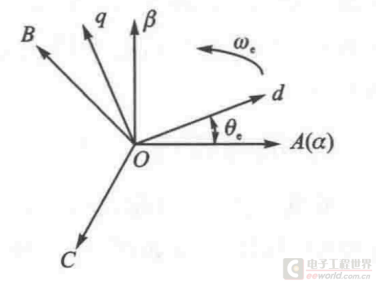


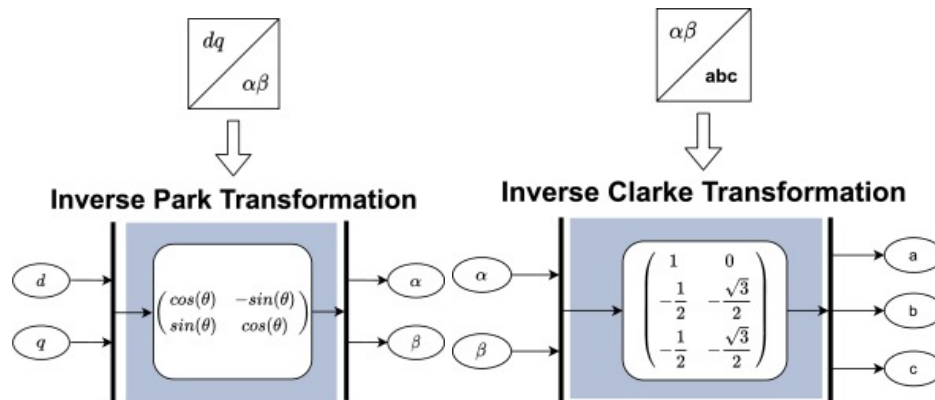
Figure 1. Field-Oriented Control scheme.

## Clarke and Park Transforms

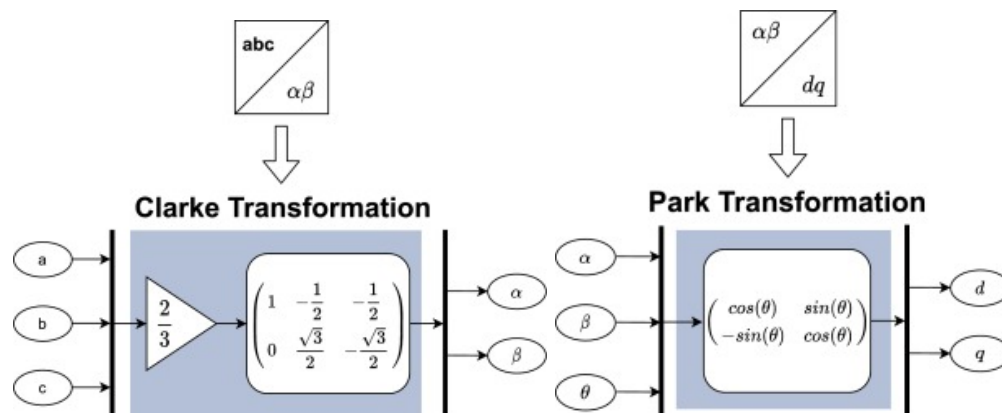
Within the FOC control loop, matrix transforms are used to convert values between torque/speed/position commands and the three phase currents in the motor windings (A,B,C) across a stationary reference  $\alpha, \beta$  coordinate system and a rotating  $d/q$  reference coordinate system.



The PI torque controllers can receive a current signal  $I_d$  and  $I_q$ , and send  $d/q$ -axis values into inverse Park and inverse Clarke transforms to produce a PWM output  $v_A, v_B, v_C$ , which enters a three-phase inverter, and to the PWM signals.



Current can be sensed for the three A, B, C phases upon exiting the three-phase inverter to be fed back into the Clark and Park transforms. The Clarke Transform converts three-phase currents  $I_A, I_B, I_C$  into a stationary  $\alpha, \beta$  coordinate system ( $I_\alpha, I_\beta$ ) and the Park Transform converts these currents into the rotating  $d, q$  axes ( $I_d, I_q$ ).



Additionally, a sensor in the form of a Hall sensor or absolute encoder provides position feedback into the Park/inverse Park transforms. Velocity data can also be read by differentiating the position information.

The current/torque loops in the PI/torque controllers run at 10kHz-20kHz, synced with the PWM switching frequency/inverter switching frequency, while the position and velocity loops run at a slower rate, from 100Hz-5kHz.